

Question 1A State Representation

[2 marks]

This is somewhat of a trick question. We just need to keep track of one integer that is the current location of the frog.

There are 4 possible actions: either 1 step or k steps forward. It is not wrong for the frog to go backwards, but it is not necessary for this question and this will create repeated states and change the answers to the later parts.

[+1] for the correct state representation.

[+1] for stating the 4 state transitions (actions).

Question 1B Initial and goal states

[2 marks]

Initial state: $p = 0$, where p is the position of the frog

Goal state: $p > n$.

[+1] for initial state.

[+1] for goal state.

Note that if (A) is blank or wrong, this question will automatically be zero marks.

Question 1C Which of the following statement(s) is/are true?

[3 marks]

- ☐ The search tree is finite.
- ☒ There are possibly many repeated states.
- ☒ There are possibly multiple goal states.
- ☐ We can always find an answer (valid sequence of moves) if we employ the right search algorithm.
- ☐ None of the above.

Question 1D Best search algorithm. Explain.

[2 marks]

BFS, because we need the minimal number of steps.
Not DFS, because no guarantee that it will be minimal.
Not DLS, because again, no guarantee that it will be minimal.
Not IDS, because while it works, We are paying extra for no good reason.

[+1] for BFS.

[+1] for saying something sensible about 2 of the algorithms.

Note that the correct answer here depends on depends on Parts (A) to (C). Saying that none of them are any good because they cannot necessarily terminate will give a small amount of credit (+1).

Question 1E State Representation for Variant #1

[2 marks]

Now we need to track 2 integers: (i) position of the frog; and (ii) number of moves left.

There are now $2k$ possible actions: up to k steps backwards or forwards

[+1] for adding the number of moves left.

[+1] for updating list of possible actions.

The right approach here is actually somewhat dependent on the search algorithm to be used. There are 2 obvious approaches: (i) track number of moves and do DFS, or (ii) do not track the number of moves and do DLS. For the former, the failure to track the number of moves will result in zero for this part and most of the subsequent parts.

Question 1F Initial and goal states for Variant #1

[2 marks]

Initial state: $p = 0$, moves = 0 Goal state: $p > n$, moves = m

[+1] for initial state.

[+1] for goal state.

Question 1G Invariant for Variant #1

[2 marks]

$\text{moves} \leq m \text{ seq}[p] \neq \text{"-"}$

Question 1H Which of the following statement(s) is/are true for Variant #1? [3 marks]

- ☒ The search tree is finite.
- ☒ There are possibly many repeated states.
- ☒ There are possibly multiple goal states.
- ☐ We can always find an answer (valid sequence of moves) if we employ the right search algorithm.
- ☐ None of the above.

Question 1I Best search algorithm for Variant #1. Explain.

[2 marks]

DLS, because we know exactly how many moves we need.
BFS, also works.
Not DFS, might not find a solution.
Not IDS, because while it works, We are paying extra for no good reason.

[+1] for DLS/BFS.

[+1] for saying something sensible.

Note that the correct answer here depends on depends on Parts (E) to (G).

Question 1J To Graph-Search or not? That is the question.

[2 marks]

It depends! If we track the number of moves, then we can do Graph Search, but if we do not track the number of moves and do DLS then we might actually need the repeated states!! The moral of the story here is that repeated states are not always bad and removing them might be harmful if people don't know what they are doing. But it really depends on the answer in Part (I).

Question 1K State Representation for Variant #2

[2 marks]

Now we need to track 3 integers: (i) position of the frog; (ii) number of moves left and (iii) amount of energy left E .

Actions remain unchanged.

[+1] for adding the number of moves left.

[+1] for updating list of possible actions.

Question 1L Which of the following statement(s) is/are true for Variant #2?

[3 marks]

- ☒ The search tree is finite.
- ☒ There are possibly many repeated states.
- ☒ There are possibly multiple goal states.
- ☐ We can always find an answer (valid sequence of moves) if we employ the right search algorithm.
- ☐ None of the above.

The answer here depends on Parts (K).

Question 1M Best search algorithm for Variant #2. Explain.

[3 marks]

Basically will need to use uniform cost search based on the energy consumption (with pruning at E) since we need to minimize energy consumption. The search tree is finite, so we will definitely be able to determine if there's not solution.

[+1] for UCS, based on E instead of m .

[+1] for saying need to cut off at E .

[+1] for saying that we can always determine if there's a solution because search tree is finite.

Note that the correct answer here depends on depends on Part (K).

Question 2A Is h_1 admissible? Explain.

[2 marks]

☐ Admissible ☒ Not admissible

Consider the case where the position of the character is at (0,0), box 1 at (1,0), box 2 at (2,0), goal 1 at (2,0), and goal 2 at (3,0). In this case, $h_1 = 2 > h^* = 1$.

Question 2B Is h_2 admissible? Explain.

[2 marks]

☒ Admissible ☐ Not admissible

Consider the case of Sokoban with N boxes and goals. In the goal state s_g , all boxes are in their respective goal positions, thus $h_2(s_g) = \frac{0}{N} = 0 \leq h^*(s_g)$.

For non-goal states, the character can push $M \leq N$ misplaced boxes to their goal positions. The minimum number of move required to move M boxes to their goal positions is 1. For M misplaced boxes, the heuristic gives $h_2(s) = \frac{M}{N} \leq 1 \leq h^*(s)$.

Question 2C Is h_A admissible? Explain.

[2 marks]

☒ Admissible ☐ Not admissible

Consider a case of Sokoban with N boxes and goals. In the goal state s_g , all boxes are in their respective goal positions, thus $h_A(s_g) = 0 \leq h^*(s_g)$.

In contrast to numbered Sokoban, at each step, the character can push at most 1 box to the goal. Thus, when there are $M \leq N$ number of misplaced boxes, at least it will take M number of steps to put all boxes to their respective goals. Thus, $\forall_{s \in S} h_A(s) \leq h^*(s)$, where S is a set of states.

Question 2D h_B admissible and consistent? Explain.

[6 marks]

☐ Admissible ☒ Not admissible

Consider the following case: $B = \{(1,0), (2,0)\}$, $G = \{(1,0), (2,0)\}$. In this case, $h_B = 1 > h^* = 0$.

☐ Consistent ☒ Not consistent

We know that consistency implies admissibility. Thus, if a heuristic is not admissible, it is also not consistent (by the contrapositive).

Question 2E Is h_C admissible and consistent? Explain.

[6 marks]

☒ Admissible ☐ Not admissible

Consider a relaxed game of Sokoban where the character can jump to any position with 0 cost and need to only push one box which has the maximum closest distance to its goal position. In this relaxed problem, the number of steps required to solve the game is exactly the heuristic cost h_C . Thus, h_C is admissible.

☒ Consistent ☐ Not consistent

h_C is the exact cost for the relaxed game of Sokoban as described above, thus h_C is consistent.

Question 2F h_D admissible and consistent? Explain.

[6 marks]

☒ Admissible ☐ Not admissible

To solve the game, the character minimally needs to move (zero the gap) to a box (which may or may not be closest), then to a goal position (which also may or may not be the closest). Thus, $\forall_{s \in S} h_D(s) \leq h^*(s)$, where S is a set of states.

☐ Consistent ☒ Not consistent

Consider the following case. We start at a state u : $p = (0, 1), G = \{(1, 0), (0, 4)\}, B = \{(1, 0), (0, 2)\}$ which has $h_D(u) = 2$. Suppose that we choose an action a to move right, then it will result in a state v : $p = (1, 1), G = \{(1, 0), (0, 4)\}, B = \{(1, 0), (0, 2)\}$ with $h_D(v) = 0$. Since $c(u, a, v) = 1$, then $h_D(u) = 2 > c(u, a, v) + h_D(v) = 1 + 0$.

Question 2G Dominant relationships between the heuristics.

[4 marks]

Which heuristic(s) does h_A dominate? Shade all that apply/ies.

☐ h_B ☐ h_C ☐ h_D ☒ None

h_B is not admissible, thus there is no dominant relationship. h_C and h_D are sometimes larger and sometimes smaller, so there is no dominant relationship as well.

Which heuristic(s) does h_B dominates Shade all that apply/ies.

☐ h_A ☐ h_C ☐ h_D ☒ None

h_B is not admissible, thus there is no dominant relationship with other heuristics.

Which heuristic(s) does h_C dominate? Shade all that apply/ies.

☐ h_A ☐ h_B ☐ h_D ☒ None

h_B is not admissible, thus there is no dominant relationship. h_A and h_D are sometimes larger and sometimes smaller, so there is no dominant relationship as well.

Which heuristic(s) does h_D dominate? Shade all that apply/ies.

☐ h_A ☐ h_B ☐ h_C ☒ None

h_B is not admissible, thus there is no dominant relationship. h_A and h_C are sometimes larger and sometimes smaller, so there is no dominant relationship as well.

Question 3A The price of a 2-room house?

[2 marks]

☐ High ☒ Low ☐ Undefined

This question tests the student's ability to reason about a decision tree.

According to the given decision tree, if the age of a house is either aged or mid, then the price is low, irrespective of the number of rooms. On the other hand, if the age is new, then the price of a 2-rooms house is low.

Question 3B What is the entropy?

[2 marks]

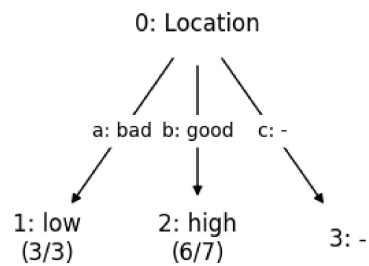
☐ $0.0 \leq \text{Entropy} < 0.2$ ☐ $0.2 \leq \text{Entropy} < 0.4$ ☐ $0.4 \leq \text{Entropy} < 0.6$ ☐ $0.6 \leq \text{Entropy} < 0.8$ ☒ $0.8 \leq \text{Entropy} \leq 1.0$

Show your working below (if you wish):

$$\begin{aligned} \text{Entropy} &= I\left(\frac{6}{10}, \frac{4}{10}\right) \\ &= -\frac{6}{10} \log_2\left(\frac{6}{10}\right) - \frac{4}{10} \log_2\left(\frac{4}{10}\right) \\ &= 0.971 \end{aligned}$$

Question 3C One-level Tree.

[6 marks]



Note: (X/Y) in the leaf nodes represent the number of samples belonging to the majority class over the total number of samples across all classes. To illustrate, a leaf node labeled as "high (6/7)" indicates that out of 7 samples, 6 of them belong to the "high" class.

What is node 0?

- | | | | |
|----------------------------|---------------------------------|---|-----------------------------|
| ☐ Age | ☐ Condition | ☒ Location | ☐ Rooms |
| ☐ High | ☐ Low | ☐ Low/High | ☐ - |

What is node 1?

- | | | | |
|----------------------------|--------------------------------------|--------------------------------|-----------------------------|
| ☐ Age | ☐ Condition | ☐ Location | ☐ Rooms |
| ☐ High | ☒ Low | ☐ Low/High | ☐ - |

What is node 2?

- | | | | |
|---------------------------------------|---------------------------------|--------------------------------|-----------------------------|
| ☐ Age | ☐ Condition | ☐ Location | ☐ Rooms |
| ☒ High | ☐ Low | ☐ Low/High | ☐ - |

What is node 3?

- | | | | |
|----------------------------|---------------------------------|--------------------------------|------------------------------------|
| ☐ Age | ☐ Condition | ☐ Location | ☐ Rooms |
| ☐ High | ☐ Low | ☐ Low/High | ☒ - |

What is edge a?

- | | | | | |
|----------------------------|---------------------------|----------------------------|--------------------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☒ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge b?

- | | | | | |
|---------------------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☒ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge c?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|------------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☒ - |

Question 3C One-level Tree. (Continued)

[6 marks]

Show your working below (if you wish):

$$\begin{aligned}
 \text{Remainder}(\text{Location}) &= \frac{7}{10} \left(I\left(\frac{6}{7}, \frac{1}{7}\right) \right) + \frac{3}{10} \left(I\left(\frac{3}{3}\right) \right) \\
 &= \frac{7}{10} \left(-\frac{6}{7} \log_2\left(\frac{6}{7}\right) - \frac{1}{7} \log_2\left(\frac{1}{7}\right) \right) + \frac{3}{10} \left(-\frac{3}{3} \log_2\left(\frac{3}{3}\right) \right) \\
 &= \frac{7}{10} (0.592) + \frac{3}{10} (0.000) \\
 &= 0.414 + 0.000 \\
 &= 0.414
 \end{aligned}$$

$$\begin{aligned}
 \text{Remainder}(\text{Condition}) &= \frac{3}{10} \left(I\left(\frac{3}{3}\right) \right) + \frac{7}{10} \left(I\left(\frac{4}{7}, \frac{3}{7}\right) \right) \\
 &= \frac{3}{10} \left(-\frac{3}{3} \log_2\left(\frac{3}{3}\right) \right) + \frac{7}{10} \left(-\frac{4}{7} \log_2\left(\frac{4}{7}\right) - \frac{3}{7} \log_2\left(\frac{3}{7}\right) \right) \\
 &= \frac{3}{10} (0.000) + \frac{7}{10} (0.985) \\
 &= 0.000 + 0.690 \\
 &= 0.690
 \end{aligned}$$

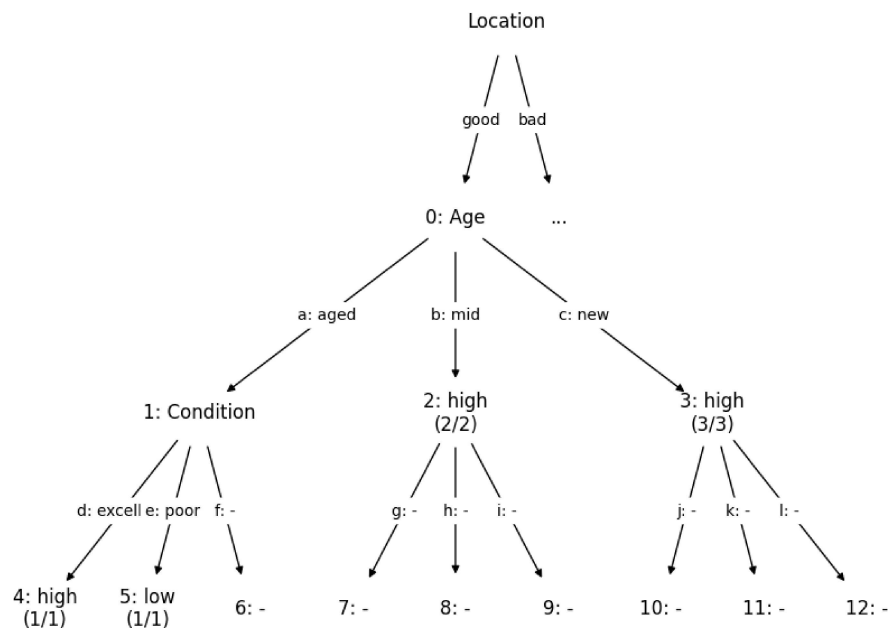
$$\begin{aligned}
 \text{Remainder}(\text{Rooms}) &= \frac{5}{10} \left(I\left(\frac{4}{5}, \frac{1}{5}\right) \right) + \frac{5}{10} \left(I\left(\frac{3}{5}, \frac{2}{5}\right) \right) \\
 &= \frac{5}{10} \left(-\frac{4}{5} \log_2\left(\frac{4}{5}\right) - \frac{1}{5} \log_2\left(\frac{1}{5}\right) \right) + \frac{5}{10} \left(-\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) \right) \\
 &= \frac{5}{10} (0.722) + \frac{5}{10} (0.971) \\
 &= 0.361 + 0.485 \\
 &= 0.846
 \end{aligned}$$

$$\begin{aligned}
 \text{Remainder}(\text{Age}) &= \frac{2}{10} \left(I\left(\frac{1}{2}, \frac{1}{2}\right) \right) + \frac{3}{10} \left(I\left(\frac{2}{3}, \frac{1}{3}\right) \right) + \frac{5}{10} \left(I\left(\frac{3}{5}, \frac{2}{5}\right) \right) \\
 &= \frac{2}{10} \left(-\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right) + \frac{3}{10} \left(-\frac{2}{3} \log_2\left(\frac{2}{3}\right) - \frac{1}{3} \log_2\left(\frac{1}{3}\right) \right) \\
 &\quad + \frac{5}{10} \left(-\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) \right) \\
 &= \frac{2}{10} (1.000) + \frac{3}{10} (0.918) + \frac{5}{10} (0.971) \\
 &= 0.200 + 0.275 + 0.485 \\
 &= 0.96
 \end{aligned}$$

We split based on "Location" as it results in the smallest remaining entropy.

Question 3D Full Decision Tree.

[6 marks]



What is node 0?

- ☒ Age ☐ Condition ☐ Rooms ☐ High
☐ Low ☐ Low/High ☐ -

What is node 1?

- ☐ Age ☒ Condition ☐ Rooms ☐ High
☐ Low ☐ Low/High ☐ -

What is node 2?

- ☐ Age ☐ Condition ☐ Rooms ☒ High
☐ Low ☐ Low/High ☐ -

What is node 3?

- ☐ Age ☐ Condition ☐ Rooms ☒ High
☐ Low ☐ Low/High ☐ -

What is node 4?

- ☐ Age ☐ Condition ☐ Rooms ☒ High
☐ Low ☐ Low/High ☐ -

What is node 5?

- ☐ Age ☐ Condition ☐ Rooms ☐ High
☒ Low ☐ Low/High ☐ -

What is node 6?

- ☐ Age ☐ Condition ☐ Rooms ☐ High
☐ Low ☐ Low/High ☒ -

Are there any nodes from 7 onwards?

- ☐ Yes ☒ No

Question 3D Full Decision Tree. (Continued)

[6 marks]

What is edge a?

- | | | | |
|---------------------------|---------------------------|---------------------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☒ aged | ☐ excellent |
| ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge b?

- | | | | |
|--------------------------------------|---------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ excellent |
| ☒ mid | ☐ new | ☐ poor | ☐ - |

What is edge c?

- | | | | |
|---------------------------|--------------------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ excellent |
| ☐ mid | ☒ new | ☐ poor | ☐ - |

What is edge d?

- | | | | |
|---------------------------|---------------------------|----------------------------|--|
| ☐ 2 | ☐ 3 | ☐ aged | ☒ excellent |
| ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge e?

- | | | | |
|---------------------------|---------------------------|---------------------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ excellent |
| ☐ mid | ☐ new | ☒ poor | ☐ - |

What is edge f?

- | | | | |
|---------------------------|---------------------------|----------------------------|------------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ excellent |
| ☐ mid | ☐ new | ☐ poor | ☒ - |

Question 3D Full Decision Tree. (Continued)

[6 marks]

Show your working below (if you wish):

$$\begin{aligned}
 \text{Remainder}(\text{Age}) &= \frac{2}{7} \left(I\left(\frac{1}{2}, \frac{1}{2}\right) \right) + \frac{2}{7} \left(I\left(\frac{2}{2}\right) \right) + \frac{3}{7} \left(I\left(\frac{3}{3}\right) \right) \\
 &= \frac{2}{7} \left(-\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right) + \frac{2}{7} \left(-\frac{2}{2} \log_2\left(\frac{2}{2}\right) \right) + \frac{3}{7} \left(-\frac{3}{3} \log_2\left(\frac{3}{3}\right) \right) \\
 &= \frac{2}{7} (1.000) + \frac{2}{7} (0.000) + \frac{3}{7} (0.000) \\
 &= 0.286 + 0.000 + 0.000 \\
 &= 0.286
 \end{aligned}$$

$$\begin{aligned}
 \text{Remainder}(\text{Condition}) &= \frac{3}{7} \left(I\left(\frac{3}{3}\right) \right) + \frac{4}{7} \left(I\left(\frac{1}{4}, \frac{3}{4}\right) \right) \\
 &= \frac{3}{7} \left(-\frac{3}{3} \log_2\left(\frac{3}{3}\right) \right) + \frac{4}{7} \left(-\frac{1}{4} \log_2\left(\frac{1}{4}\right) - \frac{3}{4} \log_2\left(\frac{3}{4}\right) \right) \\
 &= \frac{3}{7} (0.000) + \frac{4}{7} (0.811) \\
 &= 0.000 + 0.464 \\
 &= 0.464
 \end{aligned}$$

$$\begin{aligned}
 \text{Remainder}(\text{Rooms}) &= \frac{5}{7} \left(I\left(\frac{4}{5}, \frac{1}{5}\right) \right) + \frac{2}{7} \left(I\left(\frac{2}{2}\right) \right) \\
 &= \frac{5}{7} \left(-\frac{4}{5} \log_2\left(\frac{4}{5}\right) - \frac{1}{5} \log_2\left(\frac{1}{5}\right) \right) + \frac{2}{7} \left(-\frac{2}{2} \log_2\left(\frac{2}{2}\right) \right) \\
 &= \frac{5}{7} (0.722) + \frac{2}{7} (0.000) \\
 &= 0.516 + 0.000 \\
 &= 0.516
 \end{aligned}$$

We split based on "Age" as it results in the smallest remaining entropy.

If Age = "aged", then we split again:

	Rooms	Condition	Price
0	2	excellent	High
1	2	poor	Low

Based on the table, it's clear that we should split based on "Condition".

Question 3E Precision, Recall, F1.

[5 marks]

Number of True Positives (TP)?

- ☐ 0 ☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ >5

Number of False Positives (FP)?

- ☐ 0 ☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5 ☐ >5

Number of True Negatives (TN)?

- ☐ 0 ☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5 ☐ >5

Number of False Negatives (FN)?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5 ☐ >5

Precision?

- ☐ $0.0 \leq \text{Precision} < 0.2$ ☒ $0.2 \leq \text{Precision} < 0.4$
☐ $0.4 \leq \text{Precision} < 0.6$ ☐ $0.6 \leq \text{Precision} < 0.8$
☐ $0.8 \leq \text{Precision} \leq 1.0$

Recall?

- ☒ $0.0 \leq \text{Recall} < 0.2$ ☐ $0.2 \leq \text{Recall} < 0.4$
☐ $0.4 \leq \text{Recall} < 0.6$ ☐ $0.6 \leq \text{Recall} < 0.8$
☐ $0.8 \leq \text{Recall} \leq 1.0$

F1 Score?

- ☐ $0.0 \leq \text{F1 Score} < 0.2$ ☒ $0.2 \leq \text{F1 Score} < 0.4$
☐ $0.4 \leq \text{F1 Score} < 0.6$ ☐ $0.6 \leq \text{F1 Score} < 0.8$
☐ $0.8 \leq \text{F1 Score} \leq 1.0$

Show your working below (if you wish):

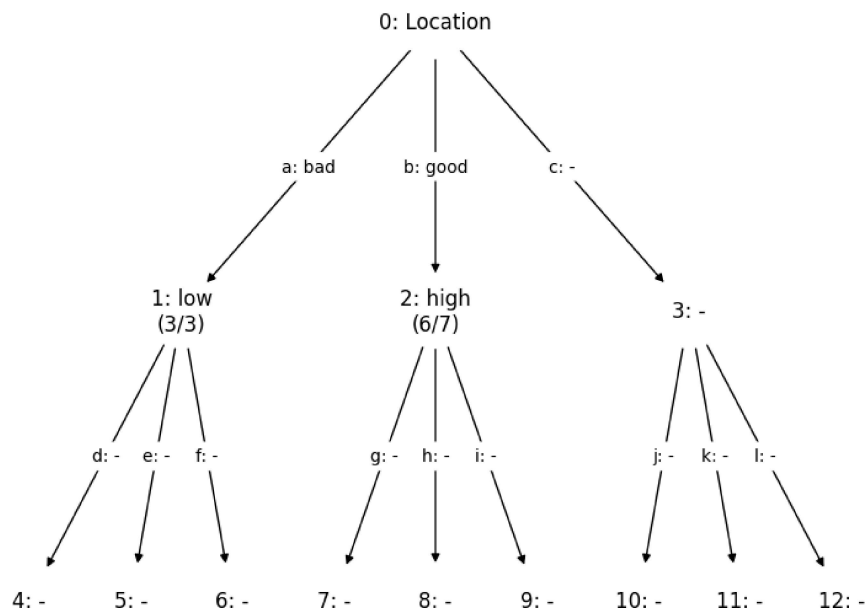
$$\begin{aligned}
 \text{Precision} &= \frac{TP}{TP + FP} \\
 &= \frac{1}{1 + 2} \\
 &= 0.333
 \end{aligned}$$

$$\begin{aligned}
 \text{Recall} &= \frac{TP}{TP + FN} \\
 &= \frac{1}{1 + 5} \\
 &= 0.167
 \end{aligned}$$

$$\begin{aligned}
 F1 &= \frac{2}{\frac{1}{P} + \frac{1}{R}} \\
 &= \frac{2}{\frac{1}{0.333} + \frac{1}{0.167}} \\
 &= 0.222
 \end{aligned}$$

Question 3F Min-Sample DT.

[3 marks]



What is node 0?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☒ Location
☐ Low/High

- ☐ Rooms
☐ -

What is node 1?

- ☐ Age
☐ High

- ☐ Condition
☒ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☐ -

What is node 2?

- ☐ Age
☒ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☐ -

What is node 3?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☒ -

What is node 4?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☒ -

What is node 5?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☒ -

What is node 6?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☒ -

Are there any nodes from 7 onwards?

- ☐ Yes

- ☒ No

Question 3F Min-Sample DT. (Continued)

[3 marks]

What is edge a?

- | | | | | |
|----------------------------|---------------------------|----------------------------|--------------------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☒ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge b?

- | | | | | |
|---------------------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☒ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge c?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|------------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☒ - |

What is edge d?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|------------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☒ - |

What is edge e?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|------------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☒ - |

What is edge f?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|------------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☒ - |

Show your working below (if you wish):

Question 4A Game Tree.

[6 marks]

What is the minimum number of variables required?

- ☒ 1
 ☐ 2
 ☐ 3
 ☐ 7
 ☐ 8
 ☐ Other

How many nodes (not including the leaves) are there in the game tree?

- ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☒ 9 ☐ 10 ☐ 11 ☐ 12 ☐ 13 ☐ 14
☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20 ☐ >20

What is the number of leaves where the first player wins?

- ☐ 0 ☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ >6

What is the number of leaves where the first player loses?

- ☐ 0 ☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ >6

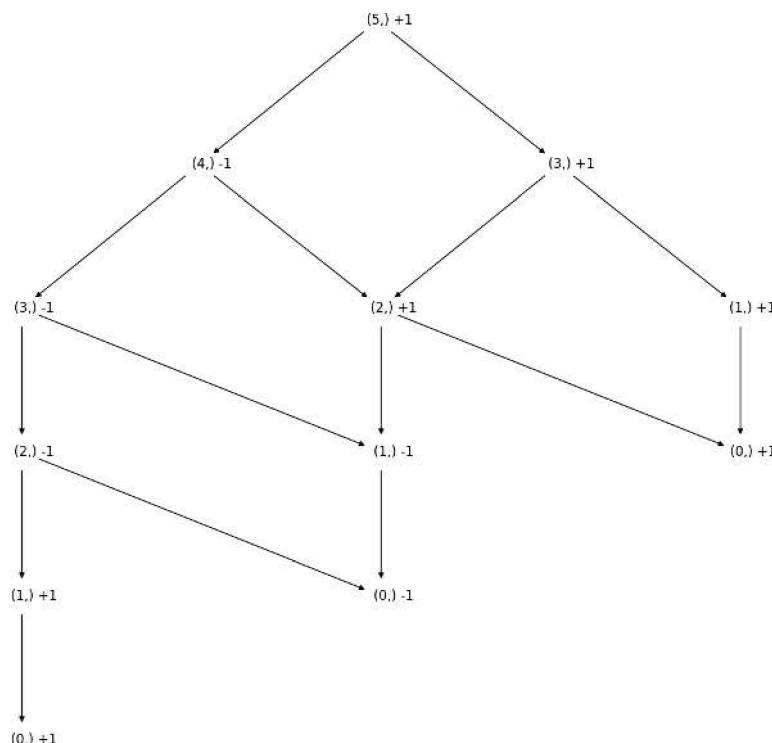
What is the total number of edges in the game tree?

- ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☐ 9 ☐ 10 ☐ 11 ☐ 12 ☐ 13 ☐ 14
☒ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20 ☐ >20

Which player do we expect to win?

- ☒ First player
 ☐ Second player
 ☐ It depends
 ☐ Draw

Show your working below (if you wish):

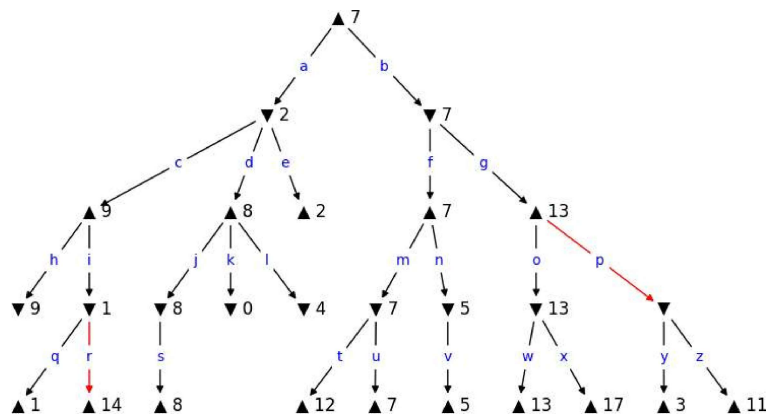


This question is here to test that the student knows how to draw a game tree and solve it correctly. This question is meant to be easy.

This is a Nim game where the starting state is the gap between two pegs.

Question 4B Alpha-beta left-to-right.

[5 marks]



What is the root value?

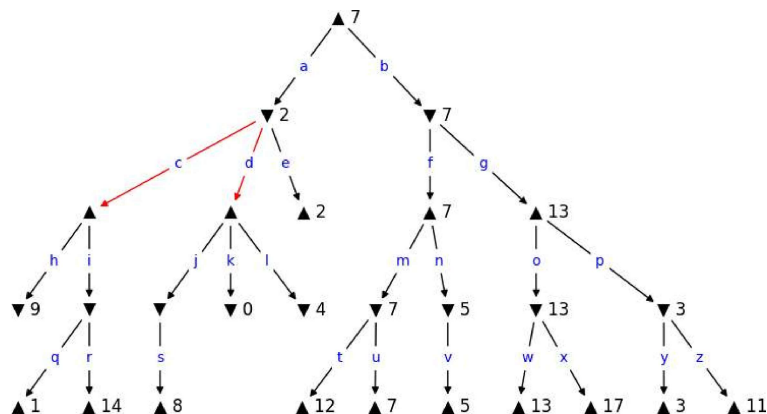
- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ 7
☐ 8 ☐ 9 ☐ 11 ☐ 12 ☐ 13 ☐ 14 ☐ 17

Which of the following link(s) are pruned? Shade all that is/are true.

- ☐ a ☐ b ☐ c ☐ d ☐ e ☐ f
☐ g ☐ h ☐ i ☐ j ☐ k ☐ l
☐ m ☐ n ☐ o ☒ p ☐ q ☒ r
☐ s ☐ t ☐ u ☐ v ☐ w ☐ x
☐ y ☐ z

Question 4C Alpha-beta right-to-left.

[4 marks]

Which of the following link(s) are pruned? Shade all that is/are true.

- ☐ a ☐ b ☒ c ☒ d ☐ e ☐ f
☐ g ☐ h ☐ i ☐ j ☐ k ☐ l
☐ m ☐ n ☐ o ☐ p ☐ q ☐ r
☐ s ☐ t ☐ u ☐ v ☐ w ☐ x
☐ y ☐ z

Question 5 3 most important lessons

[3 marks]

There are no right answers here. Students will get credit for 3 well-explained learning points that are reasonable and justified. Student needs to put in *some* effort. Clearly if the student makes a patently false statement, marks will be deducted.